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PAPER_1125: AGN Feedback, M-σ Scaling, and Metallicity Gradient Flattening in the UQFF

Abstract

Based on arXiv:2506.09123 (2025), we implement a UQFF calculator for AGN feedback effects on the $M_{\rm BH}$-σ relation and circumgalactic metallicity gradients. The classical $M_{\rm BH} \propto \sigma^{4.38}$ scaling (Kormendy & Ho 2013) is modulated by AGN feedback energy $E_{\rm AGN} = \varepsilon_f \dot{M}_{\rm acc} c^2$. High Eddington-ratio AGN flatten metallicity gradients through outflow mixing, modeled as [SCm] expulsion in the UQFF Ug4 framework with $\Delta M_{\rm BH}$ proportional to $f_{\rm feedback}$.

1. The M-σ Relation

$$M_{\rm BH} = 3.09 \times 10^8 \left(\frac{\sigma}{200\ \text{km/s}}\right)^{4.38} M_\odot$$

2. AGN Feedback Energy

$$E_{\rm AGN} = \varepsilon_f \cdot \dot{M}_{\rm acc} \cdot c^2$$

with typical radiative efficiency $\varepsilon_f \approx 0.05$ and accretion rates $\dot{M} \sim 0.01$-10 $M_\odot$/yr.

3. Eddington Ratio

$$\lambda_{\rm Edd} = \frac{E_{\rm AGN}}{L_{\rm Edd}} = \frac{\varepsilon_f \dot{M} c^2}{1.26 \times 10^{38} (M_{\rm BH}/M_\odot)}$$

4. Metallicity Gradient Flattening

AGN-driven outflows flatten the CGM metallicity gradient:

$$\nabla Z_{\rm flat} = \nabla Z_{\rm intrinsic} \cdot \frac{1}{1 + 10\lambda_{\rm Edd}}$$

High $\lambda_{\rm Edd}$ systems show nearly uniform CGM metallicity.

5. UQFF Ug4 Framework

$$f_{\mathrm{feedback}} = \varepsilon_f \cdot \lambda_{\mathrm{Edd}}$$

$$Ug4 = \rho_{\mathrm{SCm}} \cdot |\Delta M| \cdot f_{\mathrm{feedback}}$$

The [SCm] expulsion mechanism drives metal ejection proportional to how much the SMBH deviates from the M-σ expectation.

Overall alignment: **85%**.

References

- arXiv:2506.09123 — AGN Feedback and M-σ Scaling (2025).
- Kormendy, J. & Ho, L.C. (2013). ARAA, 51, 511.